Zikang Leng

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Education

Georgia Institute of Technology — Atlanta, GA
PhD in Computer Science — *Aug. 2024 - Expected May. 2028*

- Advised by Prof. Thomas Plötz. Research areas: human activity recognition, multimodal sensing, data generation, LLM agents

Georgia Institute of Technology — Atlanta, GA
B.S. in Computer Science (Theory & AI), B.S. in Physics | GPA: 4.0 — *Aug. 2021 - May. 2024*

Experience

Graduate Research Assistant — Aug. 2024 - Present
Georgia Institute of Technology — *Atlanta, GA*

- Developed **AgentSense** (AAAI '26): LLM-guided agents simulated **18** personas across **22** home layouts to generate **250 days** of privacy-preserving smart-home sensor data; pretraining on it beat baselines on **5** real HAR datasets.
- Built **EduGage** (IMWUT '26), fusing **8** synchronized physiological, motion, and eye-tracking signals from 16 participants to reach **83.18%** within-1 engagement accuracy, outperforming foundation-model and LLM baselines.
- Benchmarked virtual IMU generation against classical augmentation over **100** activities and **22** body locations, across 3 HAR benchmarks and 4 model architectures.

Machine Learning Engineer Intern — Aug. 2026 – Dec. 2026
BehaviorAI — *Brooklyn, NY*

- Developing behavior foundation models and generative methods for human behavior understanding.

Research Intern, Sensor Machine Learning — Jun. 2025 – Aug. 2025
Nokia Bell Labs — *Cambridge, United Kingdom*

- Fused IMU and PPG by signal quality at runtime for cardiovascular sensing, distilling from ECG at training.
- Self-supervised SwAV clustering for cross-modal distillation; assignment entropy gives uncertainty (**UK patent**).

Undergraduate Research Assistant — Aug. 2022 – May 2024
Georgia Institute of Technology — *Atlanta, GA*

- Developed **IMUGPT** (IMWUT '24), generating virtual IMU data from text via LLMs and motion synthesis; motion filtering and diversity metrics cut generation effort by **over 50%**.
- Evaluated alternative data-collection strategies for wheelchair transportation mode detection, reducing reliance on recruiting wheelchair users (Best Paper Nomination, ISWC '24).
- Built a personalized emotion-recognition system using wearable IMUs and virtual motion data for pretraining.
- Built **FingerSpeller** (ASSETS '23), a camera-free ASL fingerspelling system reaching **91%** character accuracy from smart rings; released a 42,288-letter dataset.

Machine Learning Research Intern – SULI — May 2023 – Aug. 2023
Argonne National Laboratory — *Lemont, IL*

- Accelerated GNN training for ocean simulation by **213**× using distributed training across **256 GPUs**.
- Expanded GNN training data via inverse-distance-weighted interpolation for larger-scale ocean simulation.

Selected Publications

IMUdrape: Generating Garment-Mounted Virtual IMU Data from Text through Physics-Based Cloth Simulation
Zikang Leng, *Thomas Plötz* — Under Review
[arxiv][code][project]

EduGage: A Multimodal Dataset and Benchmark for Sensor-Based Momentary Assessment of Engagement in Self-Guided Video Learning
Zikang Leng*, *Edan Eyal**, *Yingtian Shi, Jiaman He, Yaqi Liu, Thomas Plötz* — UbiComp/IMWUT '26
[arxiv][code][dataset][project]

AgentSense: Virtual Sensor Data Generation Using LLM Agents in Simulated Home Environments
Zikang Leng*, *Megha Thukral**, *Yaqi Liu**, *Hrudhai Rajasekhar, Shruthi K Hiremath, Jiaman He, Thomas Plötz* — AAAI '26
[paper][code][project]

*Equal Contribution

StitchSense: Smart-Garment Sensing as a Source Domain for Watch-Based Activity Recognition
*Ishita Datta, **Zikang Leng**, Rachit Bhayana, Thomas Plötz* UbiComp/ISWC '26 *[paper]*

Characterising Topic Familiarity and Query Specificity Using Eye-Tracking Data
*Jiaman He, **Zikang Leng**, Dana McKay, Johanne R Trippas, Damiano Spina* SIGIR '25 *[paper][code]*

IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition
***Zikang Leng**, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz* UbiComp/IMWUT '24 *[paper][code][project]*

More Data for People with Disabilities! Comparing Data Collection Efforts for Wheelchair Transportation Mode Detection (Best Paper Nomination)
Sungjin Hwang, **Zikang Leng***, Seungwoo Oh, Kwanguk Kim, Thomas Plötz* UbiComp/ISWC '24 *[paper]*

Emotion Recognition on the Go: Utilizing Wearable IMUs for Personalized Emotion Recognition
Zikang Leng, Myeongul Jung*, Sungjin Hwang, Seungwoo Oh, Lizhe Zhang, Thomas Plötz, Kwanguk Kim* UbiComp'24 HASCA *[paper]*

FingerSpeller: Camera-Free Text Entry Using Smart Rings for American Sign Language Fingerspelling Recognition
David Martin, **Zikang Leng***, Tan Gemicioglu, Jon Womack, Jocelyn Heath, Bill Neubauer, Hyeokhyen Kwon, Thomas Plötz, Thad Starner* ASSETS '23 *[paper]*

Generating Virtual On-body Accelerometer Data from Virtual Textual Descriptions for Human Activity Recognition (Best Paper Honorable Mention)
***Zikang Leng**, Hyeokhyen Kwon, Thomas Plötz* UbiComp/ISWC '23 *[paper][code][project][news]*

On the Utility of Virtual On-body Acceleration Data for Fine-grained Human Activity Recognition
***Zikang Leng**, Yash Jain, Hyeokhyen Kwon, Thomas Plötz* UbiComp/ISWC '23 *[paper]*

Patents

Language-based Cross-modality Transfer System for Generating Virtual Data to Enhance Activity Recognition
***Zikang Leng**, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz* US 2026/0289407 A1

A System for Quality-Aware Multimodal Fusion Representation Learning with Uncertainty Estimation for Health Condition Modelling
***Zikang Leng**, Mohammad Malekzadeh, Chi Ian Tang* UK Patent Filed '25

Awards

NSF Graduate Research Fellowship (GRFP) 2024
- Three-year $37,000 annual stipend plus $16,000 yearly tuition allowance for outstanding STEM graduate students

Georgia Tech President's Fellowship 2024
- Awarded to select doctoral students for exemplary scholarship and innovation. $5,500 annual stipend, renewable for three years

Georgia Tech Institute Fellowship 2024
- Georgia Tech's most prestigious graduate fellowship, awarded to a select few President's Fellowship holders. $2,000 annual stipend

Provost's Academic Excellence Award 2024
- Top graduating senior from Georgia Tech's College of Computing. $2,000 award

President's Undergraduate Research Awards (Salary and Travel) 2023
- $2,500 to conduct undergraduate research with Georgia Tech faculty, plus a travel grant

Technical Skills

Languages: Python, C/C++, Java, Bash, LaTeX
ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, OpenCV, Ray Tune
Infrastructure: Distributed multi-GPU training, CUDA, Docker, Git

Mentored Students

Mentored 5 masters and 5 undergraduate students; 7 are co-authors on the publications above.